

Retrieval succeeds only when five things hold, in order

The North Star is defined first, then read upward: each node below must hold for the one above it to move.

METRIC DEFINITION (PROPOSED)

$$= (\text{Retrieval attempts that start from an imprecise memory and end in a confirmed retrieval}) \div (\text{Retrieval attempts that start from an imprecise memory, in the period}) \times 100$$

● **BOTTOM-UP OUTCOME TREE · READ UPWARD**

NORTH STAR

% of imprecise-memory attempts that end in a confirmed retrieval



D5 Recover

A failed first try still ends in success, or an informed stop



D4 Recognize

User confirms the right photo, quickly and with confidence



D3 Match

The intended photo appears among the candidates returned



D2 Express

The query carries the clues the user actually holds



D1 Remember

User holds partial context — people, place, story, look, roughly when

Precondition: the photo is in the searchable library at all (Available).

- THE CEILING

Remembering is not the constraint: every one of the 800 records describes a photo the user believes exists, and 594 name a specific piece of information they lack. The constraint is what happens after the first query. The corpus is complaint-selected, so it shows where retrieval breaks, not how often it succeeds — no baseline can be computed from it, and the production value stays TBD.

● WHAT HAS TO CHANGE AFTER THE FIRST QUERY

Node	Product outcome	Diagnostic metric (proposed)
Express	Query carries the clues held	First-query clue coverage
Match	Photo is among the candidates	Target-in-candidates rate
Recognize	Confirms quickly, with confidence	Time to confirm; unsure-match rate
Recover	Failed try still ends well	Recovery rate; abandon / leave-app rate

Which node holds the leak is a research question, not an assumption: none of the 800 records states that the product misread the clues, so Match cannot be ranked from this corpus. Metrics are proposals — they need production logs.